

Jiyi Radar Custom Firmware: User Manual

PREFACE

This technical manual provides configuration and operational guidelines for integrating Jiyi CAN Radars (H30 Terrain Altimeter and R21 Obstacle Avoidance) via custom ArduPilot firmware. It contains detailed procedures for firmware deployment, hardware interfacing, and flight controller parameter configuration. Please review all instructions carefully prior to drone setup to ensure safe and reliable operation.

1 OVERVIEW

This custom firmware includes a native C++ driver designed specifically for Jiyi CAN radars. It allows the flight controller to directly communicate with the H30 and R21 sensors, providing stable, accurate, and highly sensitive distance measurements for terrain following and obstacle avoidance.

2 FIRMWARE FLASHING

The firmware is provided as a pre-compiled .apj file. Please follow the steps below to flash it to your flight controller (e.g., CubeOrange).

2.1 Locating the Firmware

1. Extract the provided .zip or project folder to your local computer.
2. Navigate to the bin folder.
3. Locate the firmware file named **arducopter.apj**.

2.2 Flashing via Mission Planner / QGroundControl

1. Open your Ground Control Station (Mission Planner or QGroundControl).
2. Connect your Flight Controller to the computer via a USB cable. **Do not connect the battery. (Do not use external power; the FC takes power from the USB cable).**
3. Navigate to the **Firmware Setup** or **Install Firmware** section.
4. Select the option for **Custom Firmware** (or "Load custom firmware").
5. A file browser will open. Navigate to the bin folder from Step 2.1 and select **arducopter.apj**.
6. Wait for the flashing process to complete and for the flight controller to successfully reboot.

For additional visual reference on flashing custom .apj files, please consult the official documentation:

- [Mission Planner Firmware Flashing Guide](#)
 - [QGroundControl Firmware Flashing Guide](#)
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3 PARAMETER CONFIGURATION

Once the firmware is flashed, you must configure the flight controller to recognize the JiYi radars.

3.1 CAN Bus Configuration

Connect the radar's CAN_H and CAN_L wires to the **CAN1** port of your flight controller. Then, set the following parameters to route the CAN bus to the Jiyi driver.

Parameter	Value	Description
CAN_P1_DRIVER	1	Enables CAN Port 1 and maps it to the first logical driver. (0 = Disabled).
CAN_D1_PROTOCOL	7	Sets the protocol to Jiyi (Mapped via the USD1 UI slot).
CAN_P1_BITRATE	1000000	Sets the CAN bus baud rate to 1 Mbps.

Using CAN2 Port (Dual CAN Flight Controllers): If your flight controller has two CAN ports and you prefer to connect the radar to the **CAN2** port, you must enable the second port and route it to the second logical driver instead. Use these parameters:

- CAN_P2_DRIVER = **2** (*Enables CAN Port 2*)
- CAN_D2_PROTOCOL = **7**
- CAN_P2_BITRATE = **1000000**

3.2 Radar Device Assignment

Configure the individual radars using the tables below.

IMPORTANT: You must enter the TYPE parameter first, **Write** the parameters, and **Reboot** the flight controller. The MSG_ID parameter will *only* appear in the list after this reboot has been completed.

Table 3-1: H30 Downward Altimeter (Rangefinder 1)

Parameter	Value	Description
RNGFND1_TYPE	37	Enables the custom Jiyi CAN driver.
RNGFND1_MSG_ID	214	H30 Target Stream ID (<i>Requires reboot to appear</i>)
RNGFND1_ORIENT	25	Downward facing.
RNGFND1_MAX	27.0	Maximum valid distance (27 meters).
RNGFND1_MIN	0.2	Minimum valid distance (0.2 meters).

Table 3-2: R21 Front Avoidance (Rangefinder 2)

Parameter	Value	Description
RNGFND2_TYPE	37	Enables the custom Jiyi CAN driver.
RNGFND2_MSG_ID	220	R21 Front Target Stream ID (<i>Requires reboot to appear</i>)
RNGFND2_ORIENT	0	Forward facing.
RNGFND2_MAX	20.0	Maximum valid distance (20 meters).
RNGFND2_MIN	0.2	Minimum valid distance (0.2 meters).

Table 3-3: R21 Rear Avoidance (Rangefinder 3)

Parameter	Value	Description
RNGFND3_TYPE	37	Enables the custom Jiyi CAN driver.
RNGFND3_MSG_ID	221	R21 Rear Target Stream ID (Requires reboot to appear)
RNGFND3_ORIENT	4	Rear / Backward facing.
RNGFND3_MAX	20.0	Maximum valid distance (20 meters).
RNGFND3_MIN	0.2	Minimum valid distance (0.2 meters).

4 VERIFICATION

To verify the radars are working correctly:

1. Power the drone (ensure the radars have power).
 2. Open the **Mavlink Inspector** or **Status** panel in your Ground Control Station.
 3. Check the DISTANCE_SENSOR messages or the sonarrange / rangefinder1 values.
 4. Move your hand or a target object in front of the radars; the distance values should update smoothly in real-time.
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5 TROUBLESHOOTING (Pre-Arm Messages)

This custom firmware includes strict safety checks. If configured incorrectly, the drone will refuse to arm and will display one of the following errors in your Ground Control Station:

- **Rangefinder X: MSG_ID is 0**
 - *Cause:* You forgot to set the MSG_ID parameter, or you did not reboot the flight controller after setting TYPE = 37.
 - *Fix:* Reboot, refresh parameters, and set the MSG_ID.
- **Rangefinder X: MSG_ID Y unsupported**
 - *Cause:* You entered an invalid CAN stream ID for the Jiyi hardware.
 - *Fix:* Ensure the ID is exactly 214, 220, or 221.